

Gaussian Quadratic Function Report

Polynomial

$$f(q) = aq^2 + bq + c$$

Naive

$$g_{\text{naive}}(X) = X^2a + Xb + c$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^2 + a\sigma^2 + bq + c$$

$$\text{Var}(g_{\text{naive}}(X)) = \sigma^2 (4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2)$$

$$\text{MSE}(g_{\text{naive}}) = \sigma^2 (4a^2q^2 + 3a^2\sigma^2 + 4abq + b^2)$$

$$\text{Bias}(g_{\text{naive}}) = a\sigma^2$$

Unbiased

$$g_{\text{unb}}(X) = X^2a + Xb - a\sigma^2 + c$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^2 + bq + c$$

$$\text{Var}(g_{\text{unb}}(X)) = \sigma^2 (4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2)$$

$$\text{MSE}(g_{\text{unb}}) = \sigma^2 (4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2)$$

Comparison

$$\text{mean gap} = -a\sigma^2$$

$$\text{variance gap} = 0$$

$$\text{MSE gap} = -a^2\sigma^4$$

$$\text{Bias}(g_{\text{naive}})^2 = a^2\sigma^4$$